

Human Centred Design

Portfolio

The design work behind BeThere, an app that turns a vague "we should hang out" into a firm plan with no organiser and no public "maybe". Grounded in a 43-response survey, three rounds of interviews, and the project's own build history.

BeThere is a group meetup-coordination app (Expo React Native client, Fastify/tRPC/Postgres backend) that turns a vague "we should hang out" into a firm plan with no organiser and no public "maybe". A creator sends one plan to a group through a single, always-anonymous flow. The group votes on candidate times and activities, then commits in a blind, deadlined moment where the only answers are Yes, Can't make it, or "I'll go if [people]". The group sees the meetup, never who proposed it.

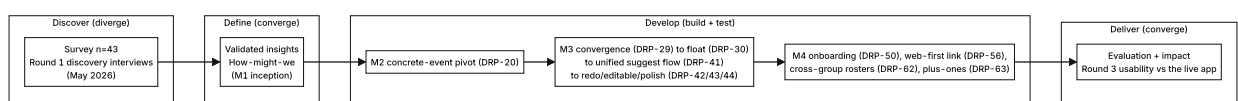
This portfolio collates the human-centred design work behind that product: the people we designed for, what we learned about them, what we built in response, and what changed as a result. It is organised around one question that the research sharpened over the term:

How might we help a group of friends turn a loose intention to meet into a committed plan, without anyone having to be the organiser and without the social pressure of a public "maybe"?

A note on evidence and honesty. Everything below traces to real primary research: a 43-response survey (collected 19 to 20 May 2026), interviews across three rounds, and the project's own Linear and git history (the design board, the cards, and the DRP issue IDs cited throughout are real). We did not run moderated, instrumented usability studies, so there is no System Usability Scale, task-success, time-on-task, or product-telemetry data. Where such a number would normally appear, it is marked [TEAM TO FILL] rather than invented, and the instrument to collect it is specified in the companion quantitative-evaluation report.

1. Process and methods

We worked the Double Diamond across the term, pairing each phase to a course milestone. Discover opened wide with the survey and a first round of discovery interviews. Define converged on the validated insights and the "how might we". Develop built and re-tested in thin slices through M2, M3 and M4, with the product genuinely changing shape in response to evidence. Deliver covers evaluation and impact.



1.1 The methods we chose, and why

Method	Scale	When	Why this method
Friend Meetup Dynamics Survey	43 respondents	19-20 May 2026	A survey tells you how widespread a behaviour is. We needed to know whether plans dying is common, and what people actually do when unsure, before committing to a build.
Round 1 discovery interviews	4 (Luca, Luke, Matthew, Noah's friend)	Pre-build	A survey cannot tell you why someone sends a "maybe". Semi-structured discovery interviews surfaced the felt mechanism behind the numbers.
Round 2 iteration usability	Tom, Felicity, Nathan, Fiza (think-aloud on the live M3 build)	M3	Reacting to a working artefact, not a description, is where the sharpest findings came from (anonymity validated, the conditional challenged).
Round 3 M4 usability	Luca (return), plus usability-test participants logged on the research board (Zack Foreman, Will Groves, Fangyi Lin, Thomas Gonzalez)	M4	To test onboarding, web-link joining and plus-ones on the real funnel, with people who had not seen the app before.

We deliberately combined a quantitative survey with three rounds of interactive interviews rather than relying on a static survey alone. The build itself was the probe: from M3 onward, participants drove a deployed product on their own phone or browser, which is what turned "we think this works" into "we watched someone use it".

1.2 How we actually ran the research

Two techniques shaped the quality of what we heard.

First, the interviews were deliberately non-leading and non-technical. The team coached each other to "sound as much as possible not like a complete CS student" and to "not use any technical words", because an earlier round had been flagged for only interviewing computing students. In the think-aloud sessions the instruction to the participant was simply "talk about everything you're thinking as you go through".

Second, we ran prototype-led research. As Lukas put it on the design board, "if we start with an implementation in code then we can use that to dig into people's brains; it worked well doing that with the mockups." Rather than ask people to imagine a flow, we deployed each interaction model and watched them use it. This is the thread that runs through Section 6: most of our design decisions were forced by a participant reacting to something concrete.

2. People - personas and stakeholders

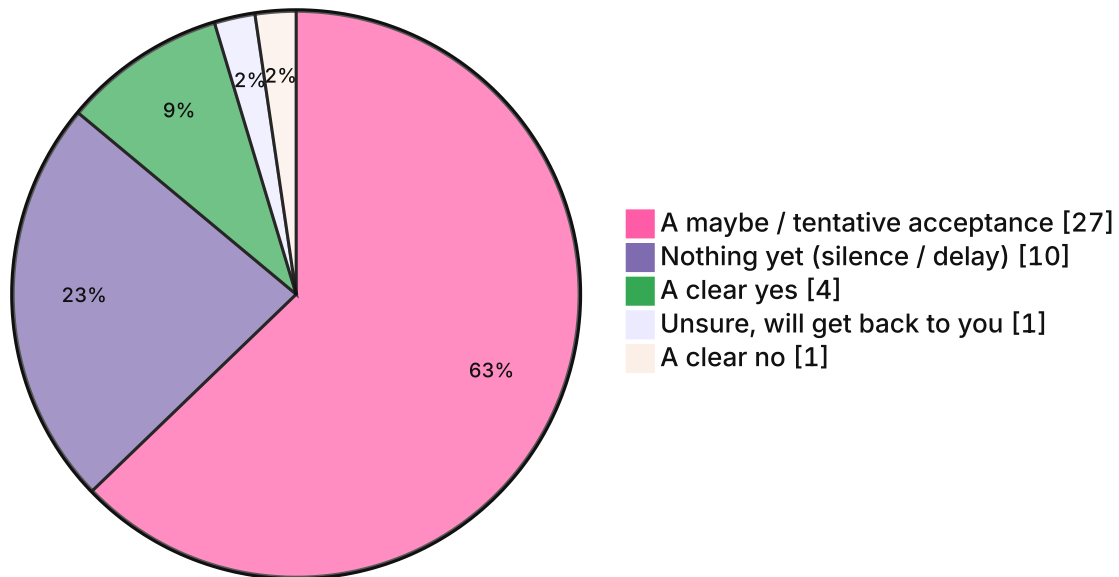
Every persona below is built from a real person in our research. Each is a composite of named participants who behaved the same way, anchored by a verbatim quote and the matching survey signal. The personas keep one truth in front of us: the plan does not die because people will not turn up; it dies in the silent gap before a time is ever fixed, and that gap is held open by the social cost of pushing and the social escape hatch of "maybe".

Note for the team: each persona should carry a real stock photograph of a person of roughly the right age, not an AI-generated face. Insert a licensed stock image in the placeholder on each persona below before submission.

2.1 How the research maps onto people

The single question that did the most to define who we are designing for: when people are only slightly unsure they can make a plan, what do they actually send back?

"When only slightly unsure, what do you send? (survey, n=43)"



37 of 43 respondents (86%) default to a maybe or to silence; only 5 send a clean answer. That is the failure mechanism in one chart, and it is why three of our four personas are defined by how they avoid committing. The fourth, the Decisive One, is the minority the product wants to make ordinary.

A second question shaped who feels the cost of coordinating: only 4 of 43 say they mostly lead, the rest watch one specific friend do it (13 of 43) or coordinate in a diffuse "evenly distributed" way (20 of 43) that, in interviews, turned out to mean nobody quite owns it. Leadership is both asymmetric and reluctantly held.

Persona	Grounded in	Defining survey signal	What BeThere must change
The Reluctant Initiator	Luca, Felicity	4/43 "mostly me"; 13/43 "one specific friend (not me)"	Make initiating cost nothing socially (anonymity)
The Hedger	Matthew, Luca	27/43 send "a maybe" when unsure	Remove "maybe"; force yes / no / conditional
The Ghost	Noah's friend	10/43 send nothing; 20/43 would use a private signal more	Make non-response safe and useful, not a dead end
The Decisive One	Tom, Luke	4/43 "clear yes", 1/43 "clear no" (the rare clean responders)	Give the deadline and push that lets decisiveness win

2.2 The personas

Persona 1 - Luca, the Reluctant Initiator

[Persona photo: real stock image of a ~20-year-old male student. Do not use an AI-generated face.]

The friend who actually wants to see everyone, and resents being the only one willing to say so.

- Goals: see his friends regularly; get a loose idea to actually happen; not feel like he is dragging people somewhere.
- Frustrations: he is usually the only one who organises; proposing feels like sticking his neck out; the group's silent default is that "it won't work out".
- What he needs from BeThere: to float an idea without it being his idea. Anonymity turns "Luca is making us commit again" into "the group is converging".

"It's almost more a lack of suggestion than the lack of execution." (Luca, Round 1)

"So I'm normally the only one who actually wants to do it." (Luca, Round 1)

Felicity sharpens the same archetype from a different group: "We have nobody who's really willing to push things... we're all quite like, oh, we'll go, but someone else has to suggest it." The Reluctant Initiator is not lazy; the act of proposing is the expensive part, and that is exactly what anonymity is designed to make free.

Persona 2 - Matthew, the Hedger

[Persona photo: real stock image of a ~20-year-old student. Do not use an AI-generated face.]

Keeps the time "semi-free" and reserves the right to ditch for something better.

- Goals: keep options open; avoid over-committing; protect a polite exit.
- Frustrations: open-ended plans drift and rot; "second thoughts" grow the longer a plan sits unresolved; he does not reliably get plans into a calendar, so he forgets.
- What he needs from BeThere: no "maybe" button to hide behind, and a deadline so the plan cannot drift into doubt.

"Maybe means I will keep the time semi-free, but if anything else comes up during that time that I feel like I would enjoy myself more, then I will [ditch]." (Matthew, Round 1)

"I think too long can also be detrimental to the success of my plan." (Matthew, Round 1)

The Hedger is the most common respondent in our data: 27 of 43 send a maybe when only slightly unsure. Luca admits the darker version: "sometimes I'm a bit more malicious when I say maybe... I don't really intend to come." Designing the maybe out of the interface is the most direct intervention BeThere makes.

Persona 3 - the Ghost (Noah's friend)

[Persona photo: real stock image of a ~20-year-old student. Do not use an AI-generated face.]

Reads the message, says nothing, and waits for someone else to commit first.

- Goals: still wants to meet (struggles even once a week); just will not move first.
- Frustrations: non-response is the failure; he waits for someone else to commit before replying.
- What he needs from BeThere: a low-cost, private way to signal interest that only surfaces if it matches others, so silence stops being a dead end.

"It's just people ignore the message. They're not responsive." (Noah's friend, Round 1)

The survey backs the need hard: 20 of 43 said they would use a private "I'd be up for something this week" signal more than messaging the group directly, with 14 more saying the same amount. For this persona,

BeThere reframes non-response from a plan-killer into a safe, private +1: you vote quietly, names are never shown, and the plan only fires if enough others matched you.

Persona 4 - Tom, the Decisive One (the minority we want to make normal)

[Persona photo: real stock image of a ~20-year-old student. Do not use an AI-generated face.]

Says yes or no cleanly, and just wants the system to make everyone else do the same.

- Goals: a clear decision, fast; the ability to commit and move on.
- Frustrations: the group will not give him a straight answer; without onboarding he was confused ("what is a group?").
- What he needs from BeThere: a deadline with a real push, plus the anonymity that makes everyone else willing to answer.

"The organisers should have an option to be anonymous... people are always really self-conscious of being the one to push things... by being able to make it anonymous there's no pressure on that person at all." (Tom, Round 2) - the single strongest validation of the anonymity model.

"I'm down if others are down." (Luke, Round 1) - the conditional that became "I'll go if [people]".

The Decisive One is rare: only 4 of 43 send a clear yes when unsure. The product's ambition is to make the rest of the group behave like Tom by removing the two reasons they do not: the cost of pushing (solved by anonymity) and the temptation to hedge (solved by removing "maybe" and adding a deadline).

A persona-level tension we are keeping honest

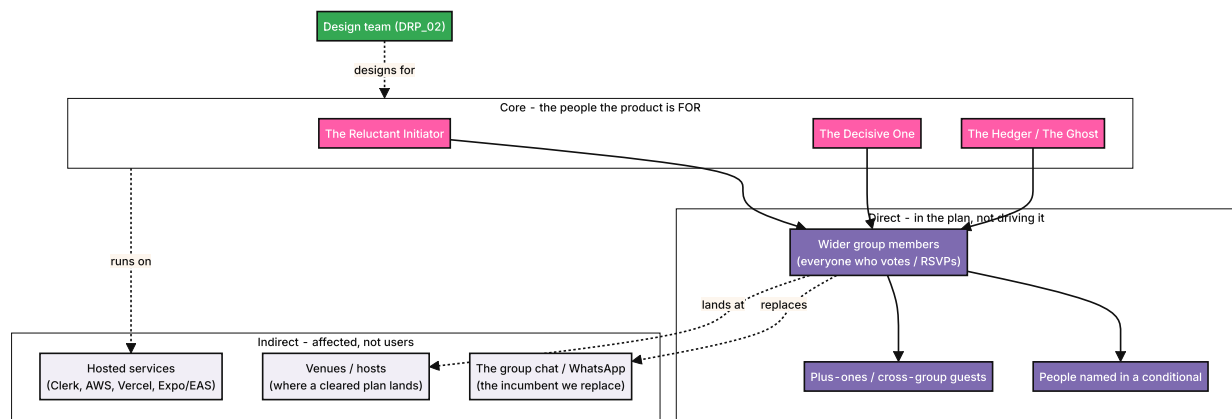
Felicity surfaced a real conflict no persona should hide. The "I'll go if [person]" conditional, even when private, felt socially impossible to her:

"If you guys are making me put something in which says I'm only going if Jess goes, I can't do that. I wouldn't be able to say it out loud. Regardless of how private it is... I would feel so awkward actually writing down which of my friends I wanted to see." (Felicity, Round 2)

Tom, by contrast, liked naming specific people. This split is a genuine, unresolved tension. We carried it forward (it directly changed the feature, see Section 6) rather than designing it away.

2.3 Stakeholder map

BeThere does not exist in a vacuum: it competes with a WhatsApp thread and depends on hosted services. The map places stakeholders in rings by how directly the product touches them.



Ring	Stakeholder	Their need	Evidence
Core	The Reluctant Initiator	Move a plan without bearing the social cost of being the pusher	4/43 lead; Luca, Felicity
Core	The Decisive One	A deadline and push so a clean yes/no closes a plan	Tom, Luke; 4/43 "clear yes"
Core	The Hedger / The Ghost	No "maybe" to hide in; a private signal so silence is not a dead end	27/43 maybe, 10/43 silence; 20/43 private - signal
Direct	Wider group members	A safe way to show interest without the maybe - cascade; public momentum without exposed names	27/43 maybe; Luke's cascade insight
Direct	People named in a conditional	Not to be put in an awkward position	Felicity's "I would feel so awkward" tension
Direct	Plus-ones / cross-group guests	To join a one-off meetup without being added to a permanent group, and not surprise the group	Luca, Zack Foreman, Nathan, Thomas Gonzalez (Round 3)
Indirect	The group chat / WhatsApp	The incumbent we must beat on convenience, or users stay put	Felicity: "Why would you bother going here if you're on WhatsApp anyway?"
Indirect	Venues / hosts	A firm time, headcount and place when a plan clears	location/notes fields feed this
Indirect	Hosted services (Clerk, AWS, Vercel, Expo/EAS)	Contractual and data - processing dependencies	See the copyright/legal report

The most important strategic stakeholder in the indirect ring is the incumbent group chat. Felicity's challenge - "it's foreign to download an app for this; I think a website would be less stressful" - is a direct shot at adoption, and it is why the project ships a web target on Vercel and a no-download share link (DRP-56). BeThere is not competing with "no tool"; it competes with the silent WhatsApp thread where, as Luke put it, when a plan dies "no one says shit".

3. Discover - research, AEIOU, insights and the opportunity

3.1 Where the problem came from

The brief was not the first idea. The team seriously explored two others before this one: a spontaneous-navigation app ("Stumble", a "Waze for walking") and a "prediction market for your friend group" where the bets friends already make about who will flake become real. The prediction-market idea was dropped on the grounds that it kept "a gambling-punishment objection" and a "comfort-zone rubric penalty", and the brief converged on meetup coordination around 19 May. The seed of the eventual mechanic appeared in Noah's discovery interview note: a friend whose meetups "fail for him about once a week, and the main reason is poor communication, people not saying whether or not they can make it", who "would wait until at least one other person gives their availability before he replies". That observation - "if only there was a discrete way to say whether or not you can make it" - is the private, conditional RSVP that BeThere is built around.

3.2 What the 43 survey responses say

Respondents are in the target group: most plan with 4 to 6 friends (25 of 43), the rest in twos and threes (13) or 7+ (4); one left it blank. The response to uncertainty is the headline: 37 of 43 (86%) default to a maybe or silence when slightly unsure (see Section 2.1). Supporting patterns, all from the same 43 responses:

- Plans die often. Asked what proportion of hangouts got talked about but never happened in the last three months, 23 of 43 said 30% or more; only 6 said under 10%.
- Bigger groups are worse. 36 of 43 (84%) said smaller groups make it easier; none said larger is easier.
- The private signal is wanted. 20 of 43 would use a private "I'd be up for something this week" signal more than messaging the group; 14 the same; 4 would use it less and 4 would not use it.
- The first time rarely sticks. Only 10 of 43 say the initial suggested time gets agreed often or always; the rest say half the time or less. This is why BeThere votes across candidate times rather than fixing one.
- Nobody wants to lead. 20 "evenly distributed", 13 "one specific friend (not me)", 5 "no one"; only 4 "mostly me".
- The hedge is real. 24 of 43 admit having lied or made an excuse to skip a planned meetup; 19 never.
- Clashing schedules is the dominant barrier (23 of 43 rate it "Often"); disagreeing on location is rare (4 of 43), which matches Luke: "99% of the time we get one place."

Bail reasons, in respondents' own words: "Couldnt be arsed", "Too tired to go potentially because its too far", "Something else came up that I wanted to go to more than the meet up", "My failure to plan accordingly, eg I forgot." The bail is usually low-stakes drift, the kind a blind, deadlined, low-pressure commitment is designed to hold together.

3.3 AEIOU

We mapped the field data through AEIOU (Activities, Environments, Interactions, Objects, Users) so the design space stays grounded in what people told us.

Lens	Observed	Evidence
Activities	Floating a vague "we should hang out"; running an informal poll and taking the majority time even if someone can't make it; postponing when it is split; chasing non-responders; forgetting the meetup.	Nathan (poll/majority/postpone); Matthew, Luca (forgetting); survey (first time rarely agreed)

Lens	Observed	Evidence
Environments	WhatsApp group chats; university term-time dispersion and academic clashes; the drawn-out limbo of an unresolved thread where second thoughts grow.	Felicity ("if you're on WhatsApp anyway"); Matthew ("days went past... drawn out... doubts creeping in")
Interactions	The "maybe" as a deliberate hedge; silence and non-response; the conditional "I'm down if others are down"; cascading uncertainty where one early "I can't" writes off the whole plan.	Luca, Matthew (maybe); Luke ("cascades uncertainty... that is a website written off"); Felicity ("fear of being the only person who says yes")
Objects	Phones; WhatsApp chats and polls; Doodle-style polls; calendars people forget to fill; the BeReal-style push people respond to.	Felicity, Tom, Nathan (polls); Matthew (calendars); Luke, Tom (BeReal-style push)
Users	Small-to-mid friend groups of 4 to 6; a single reluctant or absent organiser; members who hedge to protect themselves; people who care more about seeing each other than the activity.	Survey (4-6 dominant; lead asymmetry); Felicity ("I just want to see them")

3.4 Validated insights

Nine insights, each triangulated across the survey and interviews.

1. The bottleneck is suggestion and commitment, not execution. People who show up are fine; plans die before a time is fixed. (Luca; survey: first time rarely agreed.)
2. The "maybe" and silence are the failure mechanism, and uncertainty is contagious. (Luke: "cascades uncertainty... that is a website written off"; survey: 86% maybe or silence.)
3. Effort is asymmetric and nobody wants to be the organiser. (Felicity, Tom, Luca; survey: "one specific friend / not me".)
4. Anonymity removes the social cost of initiating; it makes a plan feel like consensus, not one person's ask. (Tom; survey: 20/43 would use a private signal more.)
5. People want to co-decide, not be organised. (Felicity: "you split the load across your friends... without feeling like you're dragging everyone".)
6. Often the company matters more than the activity. (Felicity: "I care less about what we do, I just want to see them.") So activity is optional and votable.
7. A blind, deadlined yes/no with a push beats an open-ended poll with a maybe. (Luke and Tom both reached for the BeReal-style "ding ding ding" prompt independently.)
8. Concrete and open plans are both needed; one switch should cover both. (Tom: "a checkbox or a switch... this is a concrete event you cannot change it... like a football match" - which became the lock flags.)
9. Honesty caveat: naming a friend in a conditional felt awkward even when private. (Felicity.) A genuine tension we surfaced and acted on.

3.5 The opportunity

The original framing held up under research and sharpened into the question this portfolio is built around:

How might we help a group of friends turn a loose intention to meet into a committed plan, without anyone having to be the organiser and without the social pressure of a public "maybe"?

In one line: BeThere turns a vague "we should hang out" into a firm plan with no organiser and no public "maybe" - and the research above is why every one of those words is load-bearing.

4. Current and future state - journey maps and service blueprint

The plan does not die from a logistics failure; it dies from a feelings failure. So the journey maps track emotion as the primary signal, with logistics as context.

4.1 Current state - "we should hang out" dies in a dead chat

Grounded in Matthew's bus-reunion story ("it was unclear when everyone was free... days went past... it just got more drawn out... social thoughts ended up creeping in and more and more people ended up pulling out") and Luke's cascade ("when someone expresses some kind of uncertainty, it cascades uncertainty on the group... that is a website written off"). Persona: a 4-to-6 friend group on WhatsApp.

Stage	What happens	Emotion	Pain
1. The spark	"We should do something soon!!"	Hopeful	A spark is not a plan: no time, no place, no commitment.
2. The hedge	A few "maybe!", one "what day though?", two go silent.	Guarded	The public maybe is a hedge, not an answer. Nobody has said yes.
3. The cascade	The first "ah I can't this week" lands; replies thin out.	Deflated	One early hesitation writes off the plan (Luke).
4. The drag	Days pass; nobody re-raises it; everyone waits for someone else.	Apathy	Effort is asymmetric and nobody wants to lead.
5. The doubt creep	Second thoughts settle; easy excuses surface.	Relief-tinged regret	The delay manufactures the decline (Matthew).
6. The silent death	No message ends it; the thread just stops.	Quiet disappointment	The plan fizzles in a dead chat, with a faint "who let it die?" residue.

The curve only goes down. Nothing in the chat is designed to arrest the fall: WhatsApp gives you a public maybe and infinite time, the two ingredients of the failure.

4.2 Future state - the same friends, on BeThere

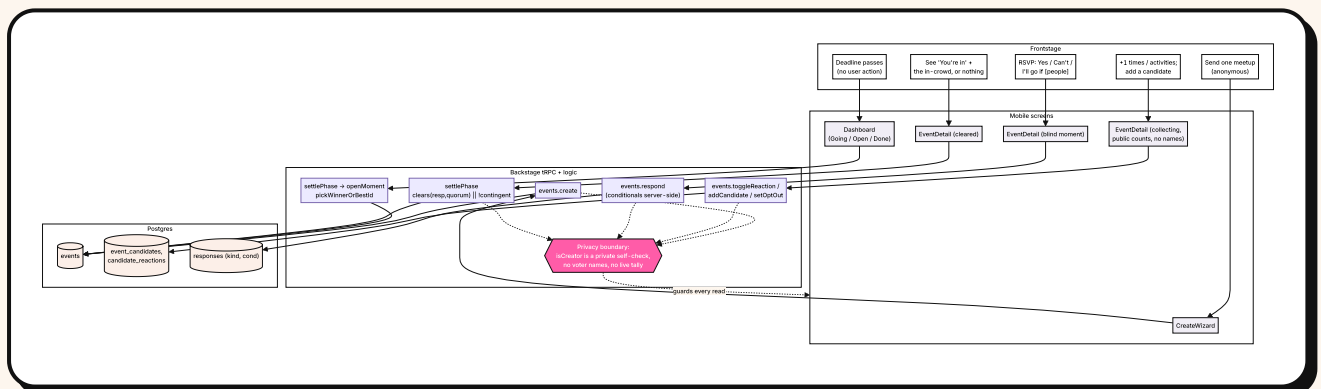
Stage	What happens	Emotion	Insight applied
1. Send (anonymous)	One person sends one meetup: candidate times, optional activities, a "Decides by" deadline. Sent anonymously.	Relieved, low-stakes	Insight 4: anonymity removes the cost of initiating. The group sees the meetup, never who sent it.
2. Collect and vote	Members +1 times and activities; anyone can add a candidate unless a lock is set. Counts show, names never do.	Engaged, safe	Insight 5: people co-decide; the load is split, with public momentum and zero exposure.
3. Momentum, not maybe	Votes accumulate. There is no "maybe" button, only a +1 or no vote.	Encouraged	Insight 2 inverted: a quiet member adds no negative drag, so one hesitation cannot cascade.
4. Auto-lock at the deadline	The top time wins automatically; the top activity becomes the plan's name if none was set.	Smooth, fair	Insight 3 + 7: no one has to call it; a deadline beats an open-ended poll.

Stage	What happens	Emotion	Insight applied
5. The blind moment	A timed moment opens; each person privately RSVPs Yes / Can't make it / "I'll go if [people]". A push nudges replies.	Decisive, a little urgent	Insight 7 + 8: no live tally and no who-is-in, so there is nothing to hedge against.
6a. It clears	At the deadline, if enough are in, it clears: "You and 4 others are in." The in-crowd is revealed only now.	Confident	The arc turns up: commitment is revealed only after it is safe.
6b. It fizzles	If not, the plan fizzles silently: no failure notice, hidden from every dashboard, no trace.	Neutral, no sting	Insight 9 / pain solved: failure leaves no dead thread and no blame.

The shape inverts the before-curve. It starts lower (a deliberate, low-stakes anonymous float) but never collapses; the deadline converts momentum into an outcome rather than letting it decay.

4.3 Service blueprint - the future flow, frontstage to database

This blueprint maps the future journey onto the real system: frontstage actions, the mobile screens (line of interaction), the backstage tRPC procedures (line of visibility), and the Postgres data. The procedure and table names are the real ones in `apps/api/src/routers/events.ts` and `apps/api/src/db/schema.ts`. A defining property sits in the backstage lane: there is no scheduler. Phase transitions settle lazily on every read and write (`settlePhase` / `openMoment`), so a plan auto-locks and clears/fizzles the next time anyone touches it, which keeps the privacy boundary inside the server.



The three privacy promises are structural, not copy. Anonymous creator: `isAnonymous` defaults true at the schema level and the read layer returns `isCreator` only as a private boolean self-check, never the creator's id. No live tally: during collecting the API returns per-candidate +1 totals but never the voter set; during the moment it returns nothing about others. Silent fizzle: a plan that misses quorum flips to `fizzled` on the next read and drops out of every dashboard, with no push and no dead thread. These guarantees are why the design's central promises live in the server's procedures, not just in the words on the screen.

5. Prototyping - the fidelity ladder

Prototyping was a deliberate fidelity ladder, not a single mock-up. We started cheap and disposable and raised fidelity only when a real question demanded it.

Rung 1 - hand-drawn M2 mock-ups (DRP-17)
Q: is the screen-to-screen flow legible?



Rung 2 - clickable walking skeleton (DRP-14/15)
real Postgres, full loop
Q: does the whole loop hold together end to end?



Rung 3 - publicly deployed live builds (DRP-16/24/31)
phone + browser, real Clerk auth
Q: will a real friend, on their own phone, get it?



Rung 4 - successive concept prototypes
concrete RSVP (DRP-20), convergence (DRP-29),
float (DRP-30), unified suggest flow (DRP-41)
Q: which interaction model do users actually want?

We climbed each rung because the one below ran out of answers. Paper (DRP-17) could tell us the journey made sense but not whether the interaction model was right, because a still image cannot show momentum counts changing or a plan silently fizzling. The walking skeleton (DRP-14/15) gave us behaviour on real Postgres, but we were still driving it ourselves. Public live builds (DRP-16 deploy on AWS App Runner + RDS, DRP-24 Clerk auth + Vercel web, DRP-31 an isolated dev deployment) removed that flattery: a friend used the actual product on their own device, which is what made the iteration interviews genuine usability tests rather than pitches. The concept prototypes (Rung 4) ran horizontally across the others: because the central question - what is the right interaction model? - could not be answered by reasoning alone, we built each candidate and tested it.

The authentic Rung 1 artefacts are the hand-drawn M2 mock-ups in `docs/mockups/m2/ALL MOCKUPS.pdf`: the create screen (which became `CreateWizard`) and the "I will make it if..." moment sheet with friends and tick boxes (the direct ancestor of the `events.respond` conditional, and of the naming tension in Section 6).

[TEAM TO FILL: built-app screenshots.] The repo ships the hand-drawn sketches but no captured screenshots of the running unified-suggest-flow build. To complete the concept-to-experience ladder, capture three screens from the live app (the create wizard, the collecting EventDetail with public vote counts, and a cleared "You're in" moment) and place them beside the sketches. They show the refined-neobrutalist visual system shipped in DRP-25.

6. Testing, iteration and analysis

BeThere was steered, not designed once. Each round tested the riskiest open assumption at that point.

Round	Hypothesis under test	Who	Method	What we looked for
R0 - Survey	Plans die in the gap before a fixed time; the mechanism is the maybe/silence, not flaking on the day.	43 respondents	Self-completion survey	A majority report plans dying; the dominant unsure reply is maybe/silence; appetite for a private signal.
R1 - Discovery	Why does the maybe cascade, and who carries the effort?	Luca, Luke, Matthew, Noah's friend	Semi-structured interviews	Causal stories explaining the survey numbers.
R2 - M3 iteration usability (DRP-9, HCD)	Does the built flow remove the organiser and the public maybe without new friction?	Tom, Felicity, Nathan, Fiza	Think-aloud on the live app	Unprompted comprehension; endorsement of anonymity; disconfirming evidence.
R3 - M4 usability	Will strangers onboard via a link and use plus-ones safely?	Luca (return), Zack Foreman, Will Groves, Fangyi Lin, Thomas Gonzalez	Think-aloud on the live web funnel	A link converts to a joined, responding member; plus-ones do not surprise the group.

6.1 Finding-to-change traceability matrix

This is the heart of the section. Every finding is real survey data or a real interview quote; every change is a real DRP issue with a verified status.

Finding (source)	Insight	Change shipped (DRP issue)
27/43 send "a maybe", 10/43 send silence when slightly unsure; only 5 a clean answer (survey).	Hedging is the default and the failure mechanism.	The moment is a blind, deadlined RSVP - Yes / Can't make it / "I'll go if [people]" - with no public maybe and no running tally (DRP-29, unified in DRP-41).
"When someone expresses uncertainty it cascades... that is a website written off" (Luke).	One early hedge collapses group certainty.	Votes and RSVPs are server-authoritative and blind during the moment; a fizzle is silent and hidden (DRP-29, DRP-41).
"The organisers should have an option to be anonymous... no pressure on that person at all... more of a consensus than one person giving a proposal" (Tom); survey: only 4/43 lead.	Anonymity reframes a plan as group consensus.	Creator anonymity is always on - the group sees the meetup, never who proposed it. No toggle to forget; it is the only mode (DRP-41).
"You split the load across your friends... suggest things without feeling like you're dragging everyone" (Felicity); 36/43 say bigger groups are harder.	People want to co-decide, not be organised.	Two public, member-fed candidate lists, TIME and ACTIVITY, that any member can add to and +1; counts public, names never (DRP-30, DRP-41).
"I care less about what we do, I just want to see them" (Felicity); activity is rarely the barrier (4/43) versus clashing schedules (23/43).	Company often matters more than the activity.	The activity is optional; the most-voted activity becomes the plan's name only if none was set (DRP-41, DRP-42).

Finding (source)	Insight	Change shipped (DRP issue)
"A checkbox or a switch... this is a concrete event you cannot change it... like a football match" (Tom); confusion over the three suggest modes.	One switch should cover concrete and open, not three modes.	<code>lockTimes</code> / <code>lockActivity</code> flags (both default open); a fully pinned plan skips collecting straight to a blind moment. The three-mode fork was deleted (DRP-41).
"The deadline would be better... but only if there was a push... like BeReal, it goes off, ding ding ding" (Tom); Luke described the same.	A blind, deadline, push-driven yes/no beats an open poll.	Lock-in deadline plus device-local "Voting closes" / "RSVP closes" reminders (DRP-32, folded into DRP-41).
The countdown bar turned greener as the deadline approached, signalling "you're nearly there, it's okay" exactly when urgency should rise (M3-review usability feedback).	Urgency was being signalled backwards.	The green drain bar was replaced with a plain-language countdown ("Voting closes in 2 days"); urgency shown in pink under one hour (DRP-43).
"It's very hard to know what days you're actually talking about... I don't even know if it's a Monday" (Felicity); "relative from today... in two days time" (Tom).	The create flow lacked everyday time scaffolding.	Day-of-week in dates, half-hour-floored default deadlines, a drawn timeline of the deadlines, and a calendar/relative picker (DRP-43, DRP-58, DRP-60, DRP-61).
"I have to put in a hard time" when you only want to float an idea (Felicity); "people are too nervous to submit an official thing" (Tom).	Loose, low-pressure suggesting must be a first-class path.	Open-by-default collecting: a plan can be sent with no lock, so the group decides time and activity collaboratively (DRP-30 float, generalised in DRP-41).
Luca could not tell which of the three suggest modes (float / rough / set) to use for a loose, see-who's-free meetup; the create page felt cluttered to Zack Foreman ("especially with the boxes").	Users think in intent, not time-precision; the form was too heavy.	The three-mode dial was deleted and the create flow became a single CreateWizard split into clear steps (DRP-41, DRP-43, DRP-44).
"Blind or raw suggesting it onto a group chat and hoping people download it is probably not it" (Fangyi Lin); "a website would be less stressful" (Felicity); "it's just a link... that feels much more seamless" (Luca, Round 3).	The download is the adoption barrier, not the idea.	Group onboarding via invite links and join codes (DRP-50) and a no-download meetup share link with a web preview that converts to a responding member (DRP-56).
"You might want to invite several groups... or a friend of a friend as a plus-one... but you don't want to invite them to the group" (Luca); Anshu's-birthday two-group case (Zack Foreman).	Real meetups cross groups and bring guests.	Freely-composed rosters: attach another group or add ad-hoc individuals by link, without polluting "My Groups" (DRP-62).
"Adding plus-ones has to be a controlled thing" (Nathan, Round 3); the failure he described is showing up to find a stranger he did not know was coming.	Plus-ones need trust controls and visibility.	A "Who's invited" roster showing who brought whom ("via X's link"), a joins-lock door, and brought-by attribution (DRP-63). Validated: "Oh that was easy" (Thomas Gonzalez); "it shows who invited him - no one's getting surprised on the day" (Luca).
"Felicity could not name friends on camera, even privately" (Round 2).	Felt privacy is not the same as technical privacy.	The conditional "Go if [people]" copy was softened and defaulted to "At least one" rather than a named person (DRP-47).

6.2 Three core interactions, shown evolving

The lecturer's guidance is to show a few core interactions changing, not every tweak. These three carried the most weight.

1. From three modes to two switches. We first built variable-precision time as a hidden branch (DRP-29), then exposed it as a Float / Rough / Set dial (DRP-30). It failed in front of a user: Luca could not map his real intent onto any of the three, and the jargon ("auto-tips", "spark", "brewing") compounded it. We deleted the dial and the entire three-mode model, replacing it with one votable plan governed by two lock flags (DRP-41). A real user simplified our information model.
2. Anonymity, from a nice-to-have to the spine. Anonymity began as one option among many. Tom's unprompted argument - "no pressure on that person at all... more of a consensus than one person giving a proposal" - and the repeated request to "be explicit about the suggestor being anonymous" hardened it into always-on, server-authoritative anonymity (DRP-41). It is the single decision that answers the asymmetric-effort insight, and it came straight from an interview.
3. The conditional RSVP, challenged and softened. The "I'll go if [people]" conditional was built on a sound insight (Luke's "I'm down if others are down"). Felicity rejected naming a friend even privately, so rather than ship it unchanged we softened the copy and defaulted to "At least one" instead of a named person (DRP-47). Felicity's stronger alternative - rank importance instead of naming - remains logged as future work.

6.3 The finding that pushed back

Not every test confirmed the design. Felicity's reaction to the conditional RSVP (Section 2.2) is the clearest disconfirming evidence we found. The privacy guarantee is real and server-resolved, but felt privacy and technical privacy are not the same thing, and one of our three Round 2 participants hit that gap hard. We recorded it, changed the default (DRP-47), and kept her ranking alternative on the table rather than averaging the tension away.

6.4 A process tension we are not hiding

One iteration (the float/wizard work) was built before its interview. Lukas pushed back on the design board: "we don't have user feedback requesting this feature... it needs a reason to be made." The team resolved to keep it but validate it properly, and agreed a rule for the future - "we've got to do interviews first" - retrofitting evidence for the one already built. This is an honest tension in our process: we mostly let evidence lead, but not always, and the discipline only held because someone in the team insisted on it.

7. Impact

We frame impact modestly. We have a 43-response survey, interviews across three rounds, and a working build; we do not have longitudinal usage data. So the claims below are the change the design is intended to produce, told through the personas, each tied to evidence that the underlying problem is real. The instruments to confirm them are specified in the quantitative-evaluation report, where every uncollected figure is marked [TEAM TO FILL].

- For Luca, the Reluctant Initiator: he can float a plan without it being his ask. Anonymity (DRP-41) means there is no one to chase and no one to blame, which directly answers the asymmetric-effort he lives with ("I'm normally the only one who actually wants to do it").
- For Matthew, the Hedger: there is no maybe to hide behind and no infinite runway. The deadline (DRP-32/41) closes the plan before his "second thoughts" can grow, the exact decay he described in the bus reunion.
- For the Ghost (Noah's friend): a quiet +1 is now a safe, useful signal rather than a dead end. Names are never shown, and the plan only fires if enough others matched him, so silence stops killing plans.
- For Tom, the Decisive One: the blind, deadlined moment with a push gives his clean yes/no a system that makes everyone else answer too.

The wider, carefully bounded claim is about friendship maintenance. Term time disperses groups, and the coordination cost of regrouping is the cost BeThere lowers. Matthew described a reunion that "got more drawn out... and more and more people ended up pulling out", and the survey shows the failure is common. If the tool helps groups clear that low-stakes hurdle, the downstream good is more in-person contact among young adults who otherwise drift apart. Felicity's line is the human core: "I care less about what we do, I just want to see them."

We do not over-claim. BeThere does not cure loneliness, and we have not measured wellbeing. The honest statement is narrower: the design removes a specific, evidence-backed friction that causes low-stakes plans to die, which should let more of them happen. The companion cover story (<docs/m4/cover-story.html>) imagines that future explicitly as an illustrative, forward-looking artefact, not a measured result.

8. Reflection

How HCD changed the product. The strongest evidence that we did human-centred design rather than feature-led design is that the product model itself changed in response to users, repeatedly. HCD deleted features (the three-mode fork, DRP-41), promoted a flag to a principle (always-on anonymity), and left a real tension on the table (naming people in conditionals) rather than pretending we resolved it. The product is smaller and sharper than the one we first imagined, and that narrowing is the clearest signal we were listening.

What we found useful and hard. Prototype-led research was the single most useful technique: deploying each interaction model and watching people use it produced sharper findings than any amount of asking. The hardest part was discipline about evidence (Section 6.4) - it is tempting to build what "just makes sense", and we did not always interview first. The finding-to-change matrix was the artefact that kept us honest, because it forced every shipped change to point back at a real quote or survey number.

What we would do differently and what is next. We would have prioritised onboarding earlier so the iteration interviews tested the whole journey, not just the core loop; onboarding only landed in M4 (DRP-50). Framed by user needs rather than a feature wishlist, the open work is: the felt-privacy problem in the conditional (test Felicity's "rank importance" alternative), reliable remote push (currently device-local only, blocked on a dev build rather than Expo Go), and the platform question of competing with the group chat people already live in (the web-first link in DRP-56 is the first move). We deliberately did not collect SUS, task-success or telemetry; DRP-57 scopes the telemetry and the quantitative-evaluation report scopes the moderated study, and we will not present an intended outcome as a measured one until those are run.

One honest bottom line: the project's own history is the best evidence of the process. A cancelled feature (DRP-40, scaling moment length by distance), a betting-market idea dropped before it began, three suggest modes collapsed into two flags, and a conditional softened after one participant could not face it - these are the marks of a design that kept changing because real people kept telling us to.